Answers
Exam in Introduction to economics for sport management

Instructions (read carefully)

a. **The final exam lasts 3 hours.**
b. **You can use calculators.**
c. **Justify all your answers using economic concepts, theories and reasoning.**

Good Luck!

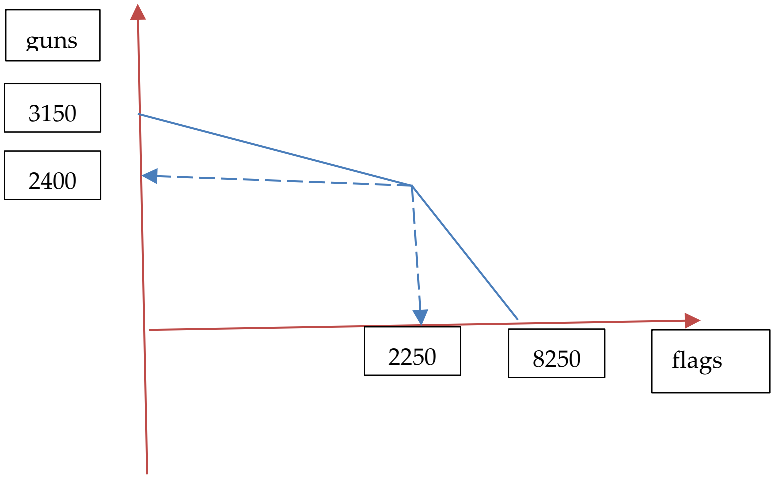
Question 1 (20 points)

Suppose there is a country that can produce two goods: flags (x) and guns (y). The country has 1200 machines and 750 workers. Each machine can produce 5 flags or 2 guns. Each worker can produce 3 flags or 1 gun.
a. Please draw a production possibilities frontier of this country and mark all the points (flags and guns) where the curve changes its slope as well as the maximal production of flags and guns. **(8 points)**
b. Suppose the country needs 750 guns. In this case, how many flags will it produce? **(3 points)**
c. Suppose there is a world trade, such that 2 flags are worth 3 guns. How many flags bad guns should the country produce and why? **(4 points)**
d. Suppose that during the world trade, the country still needs to consume the number of flags that you found in question b. In this case, how many guns will it consume? **(5 points)**

Answer:

	Machines	Workers
Production flags per one unit	5	3
Production guns per one unit	2	1
MC in producing flags in terms of guns	0.4	**0.33**
MC in producing guns in terms of flags	**2.5**	3
Max flags	1200*5= **6000**	750*3 = 2250
Max guns	1200*2 = 2400	750*1 = **750**

a. See graph below
b. 375 machines will produce 750 guns. The rest 1200-375=825 machines will produce 825*5=4125 flags. Also all the workers will produce 2250 flags. In total, there will be 2250+4125=6375 flags.
c. The country has a comparative advantage in producing flags, since its cost of production is at most 2/5, which is higher than those of producing guns. However, the country can sell flags to the world for the price of 3/2 (1.5). Therefore, the country will only produce 8250 flags and zero guns.
d. If the country needs to consume 6375 flags, then it will produce 8250 flags. The country will trade 8250-6375=1875 flags for 1875*3/2 = 2813 guns.



Question 2 (20 points)
There is a market with 6 workers who should be allocated between two fields with the following production function:

Field A

L	TP
0	0
1	10
2	19
3	27
4	33
5	37

Field B

L	TP
0	0
1	2
2	7
3	10
4	12
5	13

a. How many workers would work on each field if it is known that the price of one produced unit on field A is 1USD and the price of one produced unit on field B is 10USD? **(4 points)**
b. What would be the salary of each worker? **(3 points)**

c. What would be the profit of the owners of each field? **(3 points)**
d. Suppose a dramatic drop in demand for good that is being produced on Field B (no change in price for good produced on Field A). Explain qualitatively without calculations what will happen in the labor market in terms of migration between the fields, the salary, and Profit of the owner of Field B. **(5 points)**
e. What do you think will happen to the price of a good produced on Field B. **(5 points)**

Answer:

Field A

L	TP	mp	Vmp (p*mp)
0	0		
1	10	10	10
2	19	9	9
3	27	8	8
4	33	6	6
5	37	4	4

Field B

L	TP	mp	Vmp (p*mp)
0	0		
1	2	2	20
2	7	5	50
3	10	3	30
4	12	2	20
5	13	1	10

a. **There will be 5 workers on Field B and 1 worker on field A.**
b. **The value of the marginal product of the last worker is 10 and this will be the wage as well (VMP=W).**
c. **Profit (A)=TP*p-W*L=10*1-1*10=0**
   Profit (B)=TP*p-W*L=13*10-5*10=80
d. **There will be migration of workers from Field B to Field A. This will reduce the salary because the value of the marginal product will be lower. On the one hand the revenue will drop, but on the other hand the costs drop as well. Thus, we don't know what should happen to profit.**
e. **It is not clear because on the one hand, the demand drops, but also the supply, so it is not clear what should happen to the price of a good produced on field B.**

Question 3 (20 points)

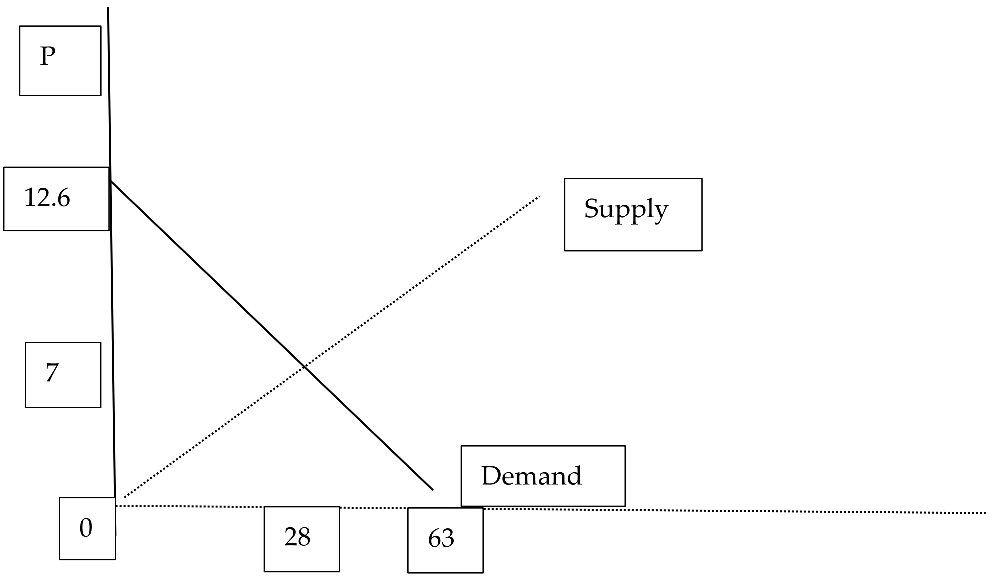
Suppose the following demand function: Q^D=63-5p. In addition suppose there are two suppliers. Each supplier has the following supply function: q^S=2p where Q^D is the quantity demanded of a good, q^S is the supplied quantity of a good of each of the two suppliers and p is the price of a good in USD.

a. What is the aggregate supply function in this market? **(3 points)**
b. What would be the price and the quantity in the equilibrium? **(3 points)**
c. Show on the graph (where appropriate) the cases where p=0, Q=0, and price and quantity in equilibrium. **(8 points)**
d. What is the producer surplus? **(2 points)**
e. What is consumer surplus? **(2 points)**
f. What is the total social welfare? **(2 points)**

Answer:

a. **The aggregate supply is** Q^S=2*2p=4p
b. **In equilibrium the supply is equal to demand (**$Q^S = Q^D$**) therefore, 63-5p=4p. Solving this equation we get p=7 and Q=28.**
c. **See the answer below**

c,d,e The total social welfare is the sum of consumer and producer surplus.
The consumer surplus is the area between the demand curve and the price (triangle area) (12.6-7)*****(28**-0)/2**=78.4
The producer surplus is the area between the supply curve and the price (triangle area) (7**-0)***(28**-0**)/2=98
The total social welfare is 78.4+98=176.4



Question 4 (20 points)

You are given the following firm's production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				50			
1	10							
2	18							
3	30							
4	45							

One unit of labor (L) costs 20$

a. Complete the table. **(10 points)**
b. What is the minimal price that the firm will produce in the short term? **(2 points)**
c. What is the minimal price that the firm will produce in the long term? **(2 points)**
d. It is known that each unit of production can be sold in the market for 240$ (p=240). What would be the production of the firm? **(2 points)**
e. What would be its profit in the short term? **(2 points)**
f. What would be its profit in the long term? **(2 points)**

Answer:

a.

TP (Q)	δQ	L	δL	MP	FC	VC	TC	MC	ATC	AVC
0	0	0	0	0	50	0	50	0	0	0
1	1	10	10	0.1	50	200	250	200	250	200
2	1	18	8	0.125	50	360	410	160	205	180
3	1	30	12	0.08	50	**600**	**650**	**240**	216.6	200
4	1	45	15	0.06	50	900	950	300	237.5	225

b. Min AVC=180. Below that price the firm will not produce in the short term.
c. Min ATC=205. Below that price the firm will not produce in the long term
d. The rule of production says that **MC= P**. If P=240$, the firm will produce 3 units of Q
e. **Profits** (short-run) =P*Q-**VC**= 240*3-**600**= 120$

   In the short run, fixed costs are sunk costs, because they are in the past and cannot be changed. Therefore, they play no role in economic decisions about future production or pricing.
   Since in the short run the profits are positive, the firm will produce.
f. Profits (long-run) =P*Q-**TC**= 240*3-**650**= 70$

   All costs are variables and therefore all the costs (fixed and variable) have to be taken into account
   Since in the long run the profits are still positive, the firm will keep producing.

Question 5 (10 points)

You were offered to invest in economic activity that yields 1000 NOK at the end of each month for three months. What is the maximal amount of money you are ready to invest in this activity if the interest rate is 8% in the first month, 6% in the second month and 5% in the third month?

Answer:

PV=1000/((1.08)(1.06)(1.05))+ 1000/((1.08)(1.06))+ 1000/(1.08)=2631

Question 6 (5 points)

A football team increased the price of its tickets. As a result, the revenue from tickets became larger. What does it say about tickets elasticity demand?

Answer:

It is inelastic (lower than 1).

Question 7 (5 points)

What would be your suggestion to decision makers if you believe that sugar drinks are not healthy, but you are unwilling to completely shut down this industry (use your knowledge from the course).

Answer:

Taxation is supposed to reduce the quantity consumed.